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CMPSC 445
Project 1 Report
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Project Description

The purpose of this project is to develop a machine learning system capable of predicting the value retention of used vehicles and identifying the factors that contribute to vehicle depreciation. It focused on collecting real world used vehicle listings from at least 2 different online automotive marketplaces,I used Cars.com and Carvana. These listings were combined with original Manufacturer's Suggested Retail Price (MSRP) reference information to calculate a value retention ratio for each eligible vehicle.

The value retention ratio was calculated using the following formula: **Value retention ratio = current listing price / original MSRP**

A higher value retention ratio usually means that a vehicle's current asking price is closer to its original MSRP, while a lower ratio implies a larger difference between the two values.

For my project, standardized base model MSRP of the vehicles was used as the reference price for each vehicle year, make, and model. So, the calculated ratio represents value retention relative to base MSRP rather than exact trim-specific depreciation.

Three regression algorithms were implemented: Ridge Regression, Random Forest Regression, and Gradient Boosting Regression. Their performances were compared using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2).

Data Collection

2.1 Data Sources

Used-vehicle listings were collected from two independent automotive marketplaces:

- **Cars.com:** https://www.cars.com/
- **Carvana:** https://www.carvana.com/

Python-based web-scraping scripts were developed to collect vehicle listings and extract relevant information from each website. Separate collection scripts were used because the websites have different page structures and provide different vehicle attributes.A API key from Scrapfly to ensure scrapping was possible through each website's security.

The project focused on three manufacturers: Honda, Nissan, and Toyota. Three models were selected for each manufacturer.

Manufacturer	Selected Models

Honda	Civic, Accord, CR-V
Nissan	Altima, Sentra, Rogue
Toyota	Camry, Corolla, RAV4

This provided a total of 9 vehicle models across three manufacturers

2.2 Number of Collected Listings

The data collection process produced a total of **1,832** results:

Cars.com = 1,119

Carvana = 713

2.3 Original Vehicle Attributes

The integrated dataset contained original listing attributes including vehicle year, make, model, trim, listing price, mileage, description, exterior color, VIN, title, condition, and SKU.

2.4 MSRP Collection

Original MSRP information was collected from automotive manufacturer pricing references and other recorded source materials.

MSRP records were standardized and matched with the used-vehicle dataset using vehicle year, make, and model. The matching process selected a standard base-trim MSRP and standard drivetrain reference.

The MSRP matching process produced this:

MSRP Matching Result	**Count**
Total collected listings	1,832
Available MSRP reference records	108
Base MSRP configurations selected	25
Successfully matched listings	1033
Unmatched listings	799

Matching rate	56.39%

Listings without an available base MSRP reference were excluded from the final modeling dataset.

Data Preprocessing and Integration

3.1 Dataset Integration

The collected Cars.com and Carvana datasets were combined into a unified dataset using Python and Pandas.Since the 2 sources used different listing structures, the extracted information was organized into a consistent set of column names. Common attributes such as year, make, model, price, and mileage were standardized before it was integrated..A source identifier was retained for each listing so that the origin of each record could be identified during later analysis.

3.2 Data Cleaning

The preprocessing pipeline included several cleaning procedures.

First, price, mileage, and MSRP values were converted into numeric formats to support mathematical calculations and regression modeling.Secondly, records with invalid or missing required numerical information were excluded from the modeling dataset. Third, duplicate detection was performed using listing URLs and valid VIN identifiers within their respective sources. .Lastly, missing attributes were handled through the preprocessing pipeline.

3.4 Preprocessing Discussion

Preprocessing was important because inconsistencies between the two websites could negatively affect model performance. Standardizing attribute names and numerical formats allowed the listings to be combined into a single dataset. A limitation was the uneven representation of manufacturers. The final dataset contained 478 Honda listings, 479 Nissan listings, and 76 Toyota listings. As a result, the trained models had substantially more examples from Honda and Nissan than from Toyota, potentially limiting the reliability of Toyota-specific predictions.

Feature Engineering

Feature engineering was performed to create additional variables that could help the regression models identify patterns associated with vehicle value retention.

4.1 Vehicle Age

Vehicle age was included because older vehicles may have experienced more wear, technological obsolescence, and depreciation than newer vehicles.

4.2 Mileage Per Year (MPY)

MPY measures the approximate annual usage of a vehicle.Two vehicles of the same age may have significantly different mileage. Mileage per year provides additional information about vehicle usage relative to age.

4.3 Features Extracted from Descriptions

Text processing was applied to vehicle descriptions to identify relevant phrases.

The following description-derived features were created:

One Owner, Accident free, Certified and AWD

The final models used the certified and AWD indicators because these features contained nonzero observations.

Certified vehicle mentions may provide information about vehicle marketing or certification status. AWD mentions may provide information about drivetrain characteristics that could affect resale value.

However, these features identify whether particular terms appeared in the listing descriptions. The absence of a term does not prove that a vehicle lacks that characteristic.

4.4 Categorical Encoding

Machine-learning models require numerical input, so categorical attributes were transformed using encoding techniques.One-hot encoding was applied to vehicle make, model, and source. This created numerical indicator columns representing the categorical values without imposing an artificial numerical ranking.Frequency encoding was applied to vehicle color. Under frequency encoding, each category is represented by its frequency in the training data:

Frequency encoding can be useful for categorical attributes with many distinct values because it does not create a separate binary column for every category.

4.5 Feature Engineering Discussion

Feature engineering allowed the models to use information beyond the original scraped attributes.

Vehicle age and mileage per year provided additional measures of vehicle usage. Description-derived indicators captured limited information from unstructured text. Categorical encoding allowed the algorithms to process manufacturer, model, source, and color information.

The final feature set included vehicle age, mileage, mileage per year, certified mention, AWD mention, make, model, source, and color.

The model year was removed from the final predictors because it was redundant with vehicle age. Condition was also excluded because it contained only one observed category in the matched dataset.

Model Development

5.1 Training and Testing Strategy

The final dataset was divided into training and testing subsets using an 80/20 split.

Dataset	**Number of Records**
Training	826
Testing	207
Total	**1,033**

The training dataset was used to fit the regression models, while the testing dataset was reserved for evaluating predictive performance on unseen records. The same training and testing partitions were used for all three algorithms to support a consistent comparison.

5.2 Ridge Regression

Ridge Regression is a linear regression method that adds an L2 regularization penalty to reduce excessively large coefficients.It was included as a regularized linear baseline for comparing against more complex nonlinear methods.The implementation used an alpha value of 10.

5.3 Random Forest Regression (RFR)

Random Forest Regression is an ensemble-learning algorithm that combines predictions from multiple decision trees.Each tree learns relationships between the input features and the target variable. The final prediction is obtained by averaging the predictions across the trees .It was

selected because it can capture nonlinear relationships and interactions between variables such as mileage, age, manufacturer, and model.

5.4 Gradient Boosting Regression (GBR)

GBR is an ensemble-learning technique that builds decision trees sequentially.

Each new tree attempts to reduce the prediction errors made by the previous trees, allowing the model to learn increasingly complex patterns. It was selected because vehicle depreciation may involve nonlinear relationships and interactions between multiple characteristics.The implementation used 200 estimators, a learning rate of 0.05, and a maximum tree depth of three.

5.5 Model Development Discussion

The three algorithms were selected to compare different approaches to regression.

Ridge Regression provided a relatively simple linear baseline. Random Forest provided a nonlinear ensemble method based on averaging decision trees. Gradient Boosting provided a sequential ensemble approach that attempts to improve predictions by correcting earlier errors.

Using three different algorithms made it possible to determine whether more flexible nonlinear models improved prediction accuracy for the collected vehicle data.

Evaluation Results

Three evaluation metrics were used to compare model performance.

6.1 Mean Absolute Error

Mean Absolute Error measures the average absolute difference between actual and predicted values:

Lower MAE values indicate more accurate predictions.

6.2 Root Mean Squared Error

Root Mean Squared Error measures prediction error while assigning greater penalties to larger errors:

Lower RMSE values indicate better performance.

6.3 Coefficient of Determination

Higher R^2 values generally indicate better predictive performance.

6.4 Model Comparison Results

Regression Model	MAE	RMSE	R^2
Ridge Regression	0.0626	0.0846	0.6772
Random Forest	0.0563	0.0734	0.7572
Gradient Boosting	**0.0546**	**0.0724**	**0.7637**

Gradient Boosting achieved the best overall performance, producing the lowest MAE and RMSE and the highest R^2 score. Its R^2value of 0.7637 indicates that it explained ~76.37% of the variation in the base-MSRP-relative retention ratio on the test dataset. The MAE of 0.0546 represents an average absolute error of ~5.46 percentage points in the predicted retention ratio.Random Forest achieved similar performance, with an R^2 value of 0.7572. Ridge Regression produced the lowest R^2score of 0.6772.

6.5 Evaluation Discussion

The results indicate that the nonlinear ensemble methods performed better than the regularized linear model on this dataset. One possible explanation is that vehicle value retention depends on interactions between manufacturer, model, mileage, and age that are not adequately captured by a simple linear relationship.Gradient Boosting may have performed better because it progressively reduced prediction errors while learning nonlinear relationships.However, the difference between Gradient Boosting and Random Forest was relatively small. Gradient Boosting had an RMSE of 0.0724 compared with 0.0734 for Random Forest.Therefore, although Gradient Boosting was the best model in this experiment, the results do not establish that it would always outperform Random Forest on different datasets or future vehicle listings.

Depreciation Analysis and Feature Ranking

To identify the most influential predictors, permutation feature importance was calculated using the best-performing Gradient Boosting model.Permutation importance measures how much model performance decreases when a feature's values are randomly shuffled.The importance values below represent the average increase in test RMSE caused by shuffling each feature. Larger positive values indicate greater predictive importance.

Rank	Feature	Importance

1	Make	0.051485
2	Mileage	0.040486
3	Model	0.039180
4	Mileage per year	0.012368
5	Vehicle age	0.004904
6	Source	0.001164
7	Color	0.000523
8	Certified mention	0.000507
9	AWD mention	0.000266

Depreciation Analysis Discussion

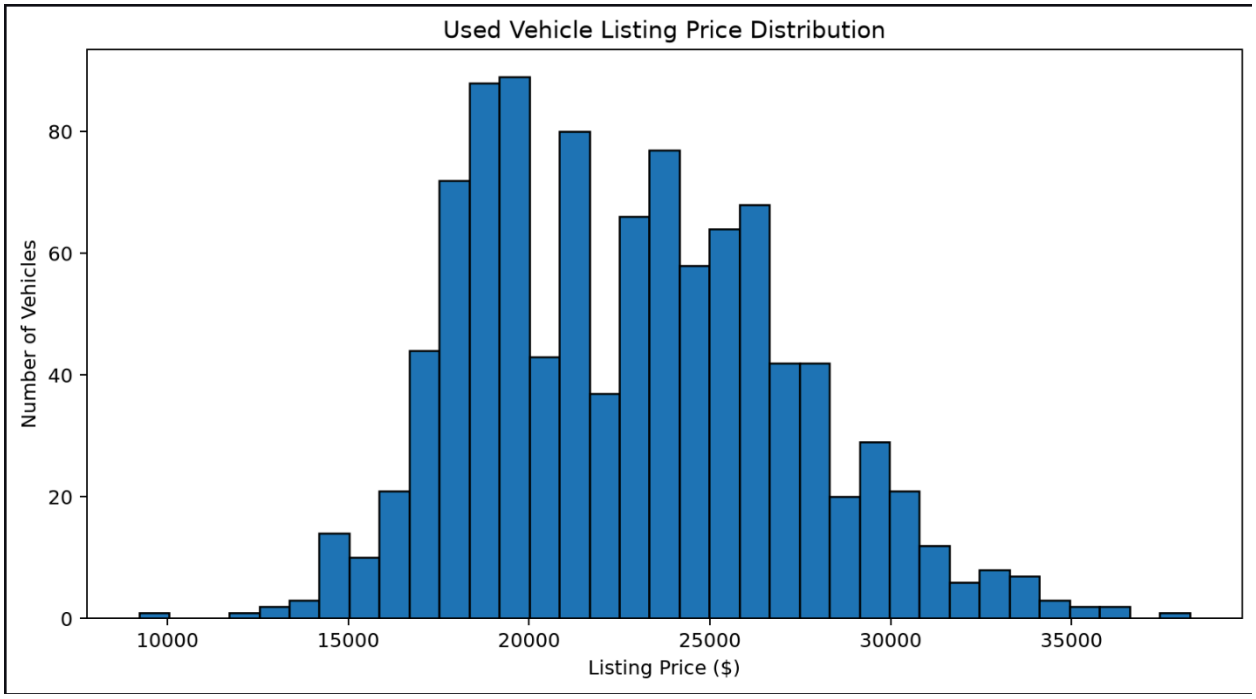
The strongest predictors were make, mileage, and model. Those specific attributes prove that vehicle value retention is influenced by both vehicle identity and usage-related characteristics. However, the analysis measures associations in advertised listing prices, not confirmed resale transactions. Furthermore, the standardized base MSRP denominator may influence differences across vehicle models.The results therefore support conclusions about predictive patterns within the dataset rather than definitive causal claims about depreciation.

Visualizations and Discussion

Several visualizations were generated using Python and Matplotlib to explore the dataset, compare models, and interpret feature importance.

8.1 Vehicle Price Distribution

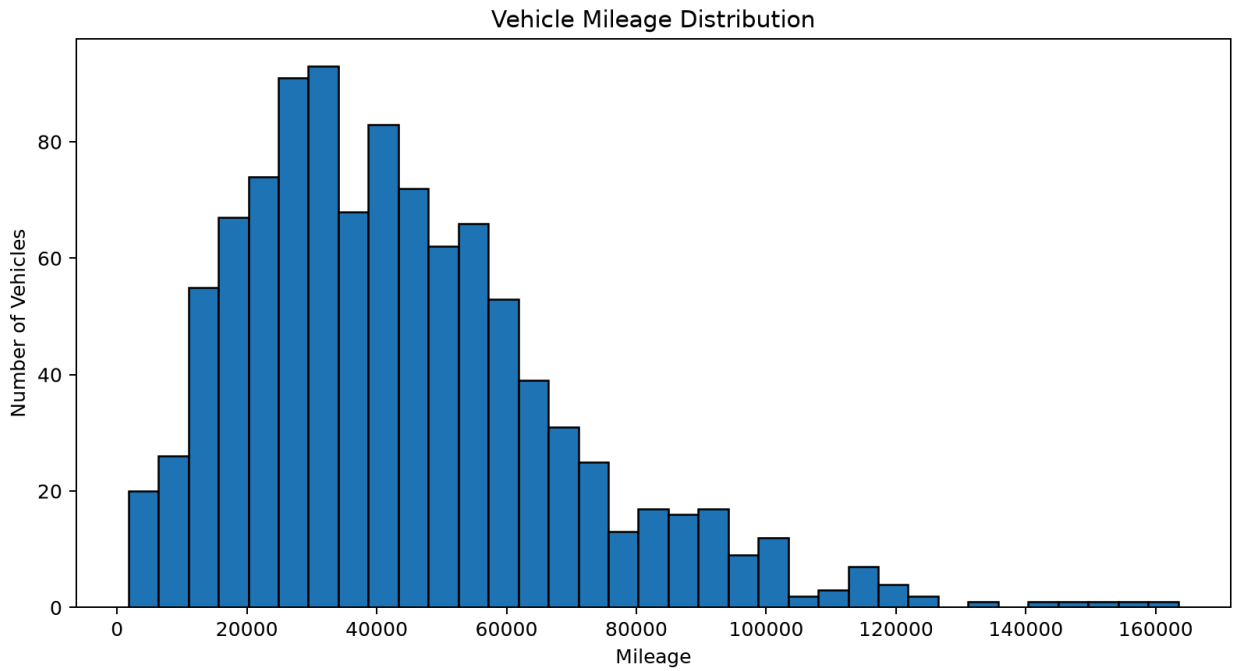
Figure 1: Used Vehicle Price Distribution



The price distribution visualization illustrates the range and frequency of used-vehicle listing prices.The final dataset had an average listing price of ~$22,790 and a median price of $22,590.These values indicate that the dataset was concentrated around the low-to-mid-$20,000 price range, although the listings included both lower-priced and higher-priced vehicles.Understanding the price distribution is useful because it provides context for the vehicle market represented by the collected data.

8.2 Mileage Distribution

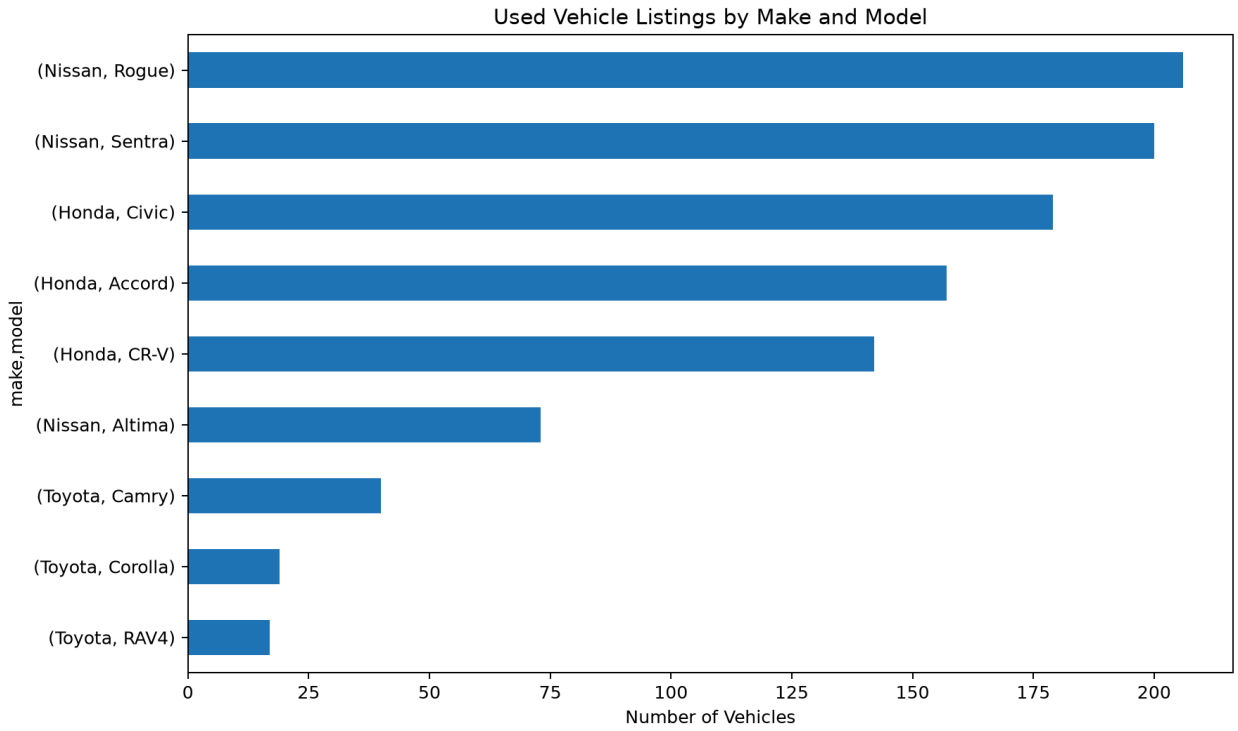
Figure 2: Vehicle Mileage Distribution



The mileage distribution visualization shows the variation in vehicle usage across the dataset.The average mileage was approximately 43,839 miles, while the median was 39,989 miles.Mileage ranged from approximately 1,752 miles to 163,455 miles.This variation provided useful information for training the regression models because vehicles with different mileage levels may have different resale values.

8.3 Manufacturer and Model Frequencies

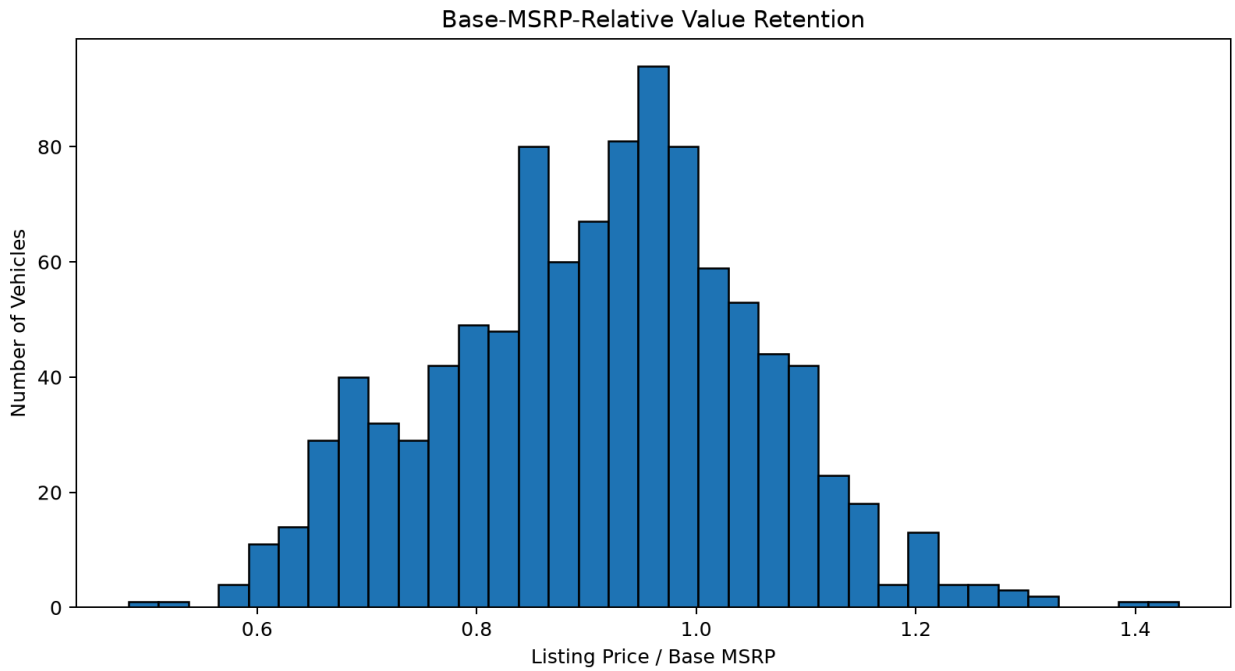
Figure 3: Manufacturer and Model Frequencies



The manufacturer and model frequency visualization shows the number of matched vehicle listings available for each selected model.The final dataset contained 478 Honda listings, 479 Nissan listings, and 76 Toyota listings.This visualization is important because it reveals the uneven distribution of the dataset. Honda and Nissan were represented much more heavily than Toyota.The imbalance may affect how reliably the models generalize to different manufacturers.

8.4 Value Retention Distribution

Figure 4: Value Retention Ratio Distribution



The value retention distribution illustrates the relationship between used listing prices and standardized base MSRP references.The average retention ratio was ~0.9121, and the median was ~0.9221.A total of 280 listings had retention ratios above 1.0.These observations do not necessarily represent errors. Higher-trim vehicles may have current listing prices above the original base-trim MSRP used as the denominator. Other factors, such as pricing differences and market conditions, may also contribute.This reinforces the importance of interpreting the ratio as base-MSRP-relative retention rather than exact vehicle depreciation.

8.5 Model Performance Comparison

Figure 5: Regression Model RMSE Comparison

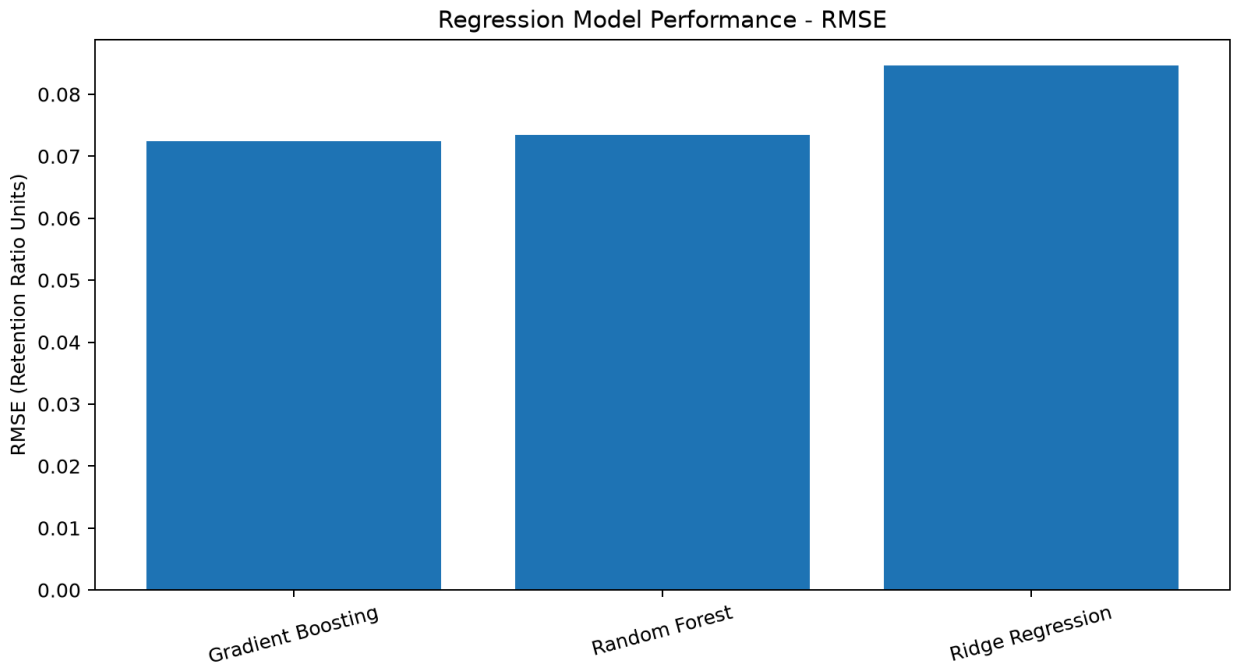
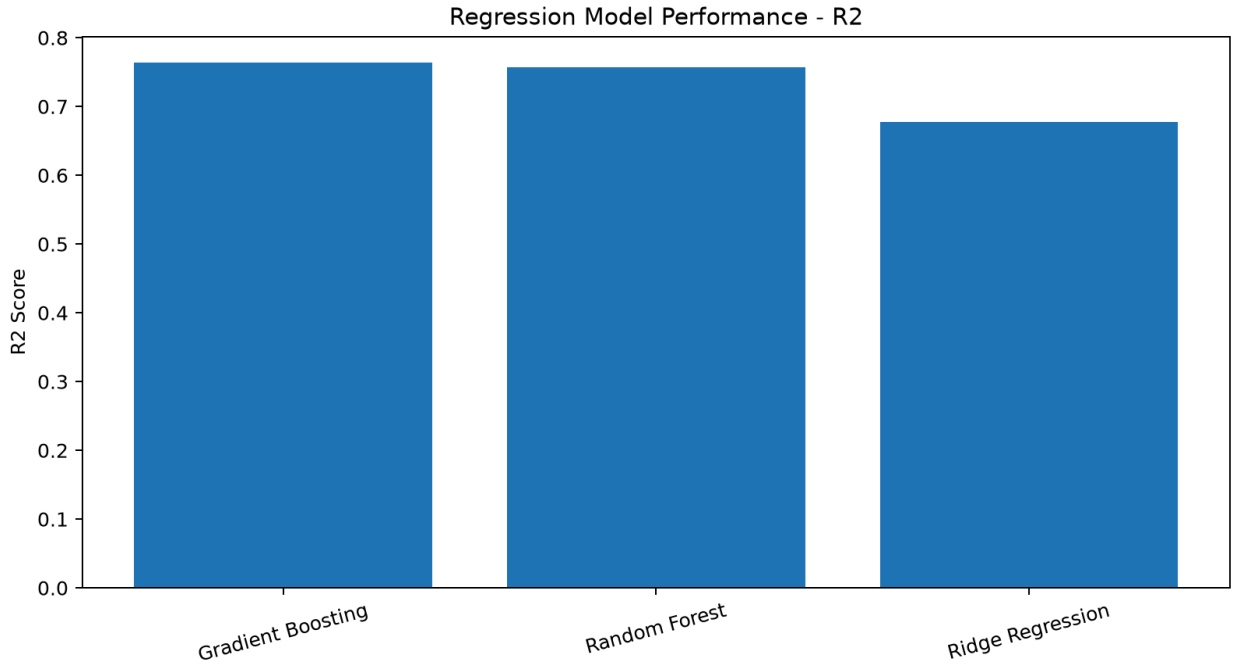


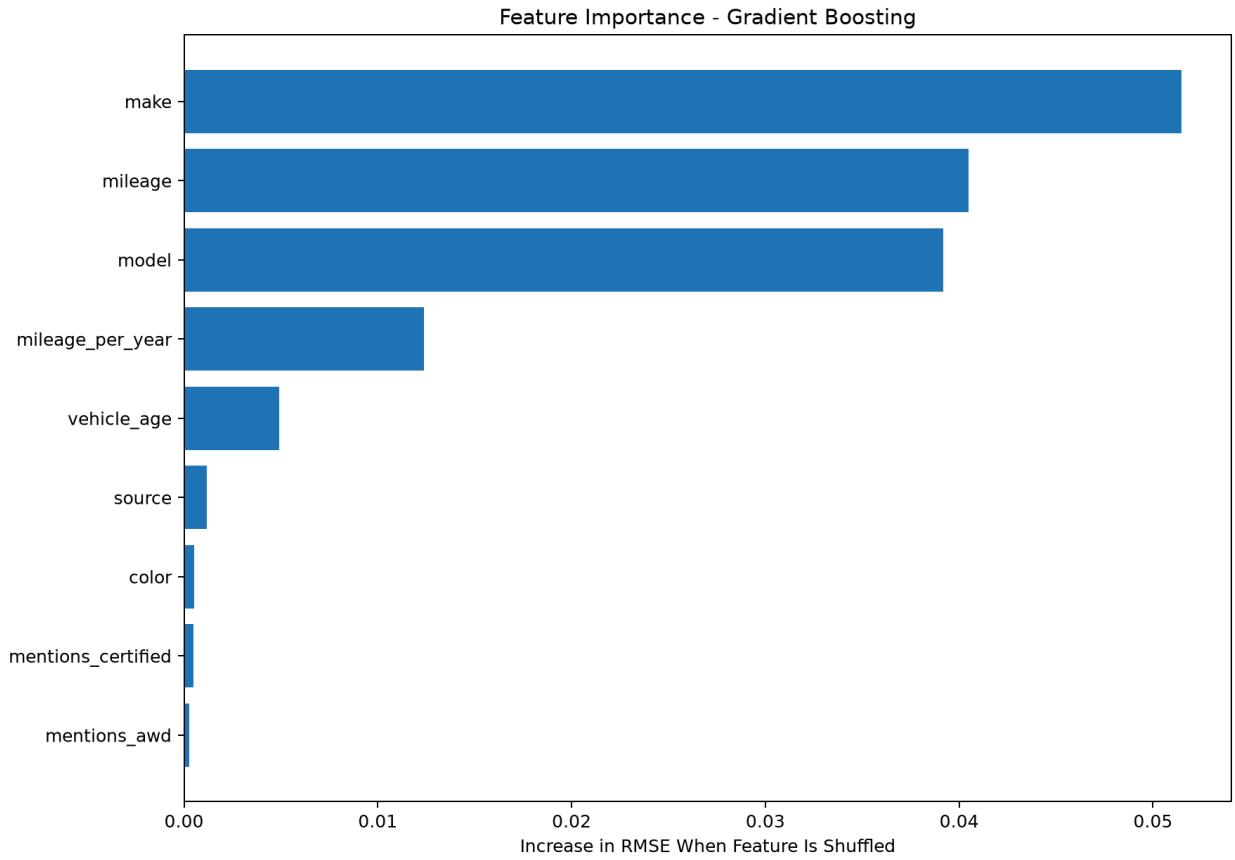
Figure 6: Regression Model R^2Comparison



The model comparison visualizations demonstrate that Gradient Boosting achieved the lowest prediction error and highest R^2score.Random Forest performed similarly, while Ridge Regression had higher errors.These results suggest that nonlinear ensemble methods were better suited to the relationships represented in the collected vehicle data.

8.6 Feature Importance

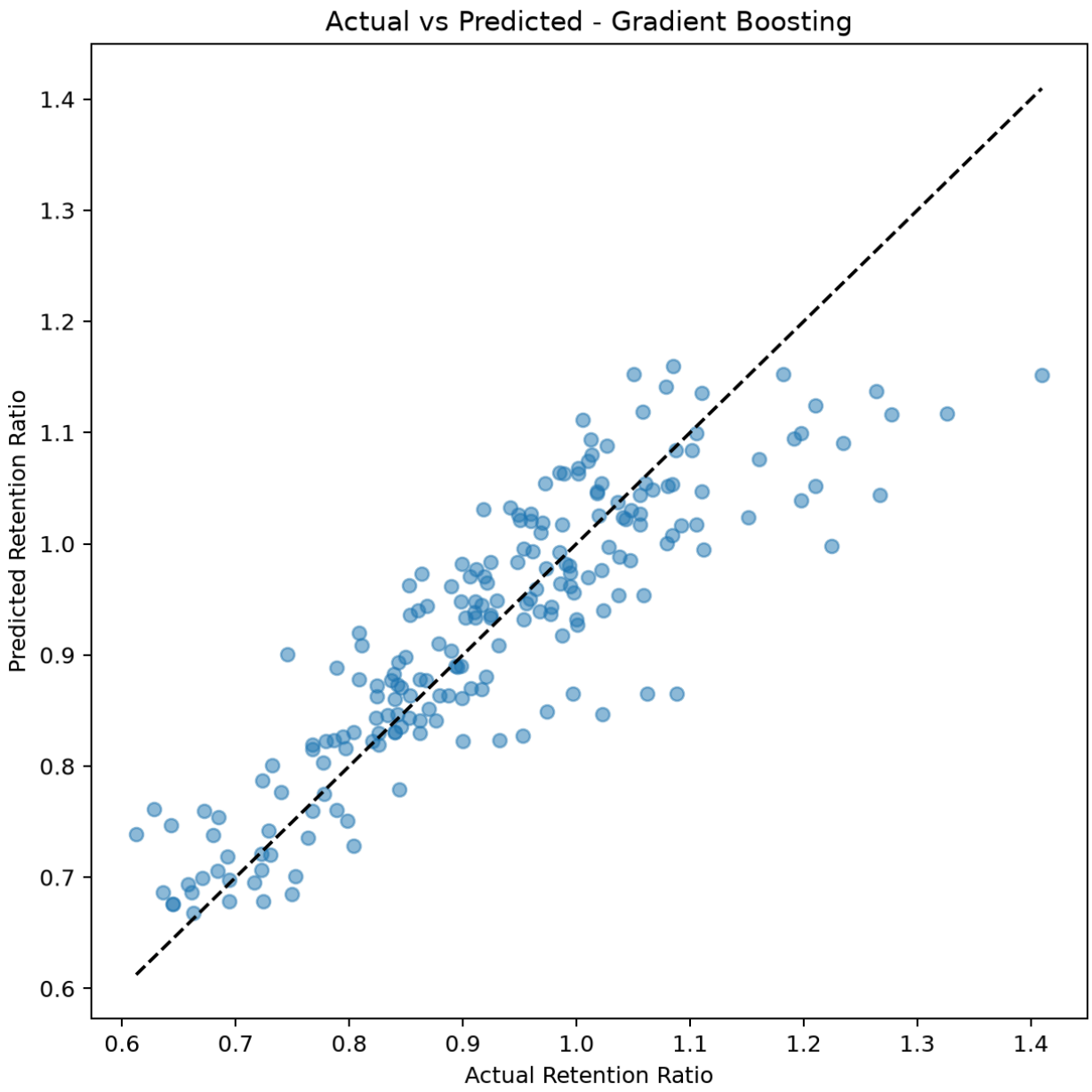
Figure 7: Permutation Feature Importance



The feature importance visualization identifies the predictors that had the greatest effect on Gradient Boosting's test performance.Vehicle make, mileage, and model were the most important features.The chart supports the conclusion that vehicle identity and usage information were central to predicting the retention ratio.

8.7 Actual Versus Predicted Retention

Figure 8: Actual Versus Predicted Value Retention



This visualization compares the actual retention ratios with predictions from the best-performing model.A prediction close to the actual value appears near the ideal diagonal relationship.The chart provides a visual way to examine model accuracy and identify instances where the model overestimated or underestimated retention.

8.8 Depreciation and Vehicle Usage

Figure 9: Value Retention by Vehicle Age

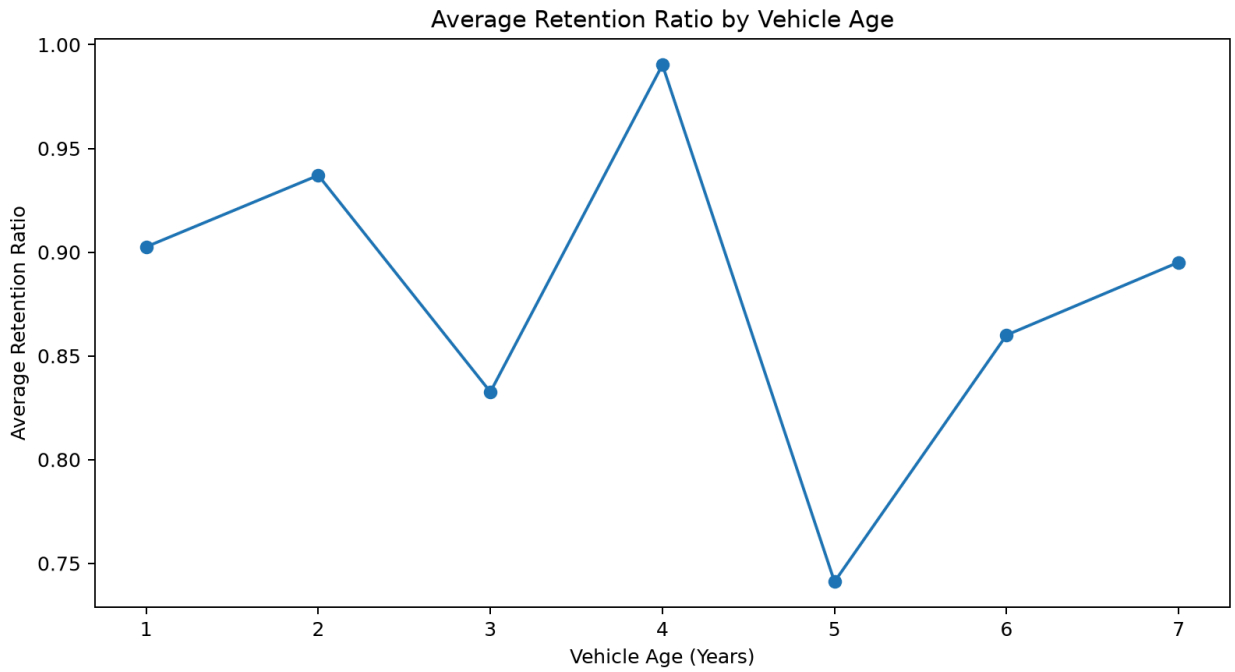
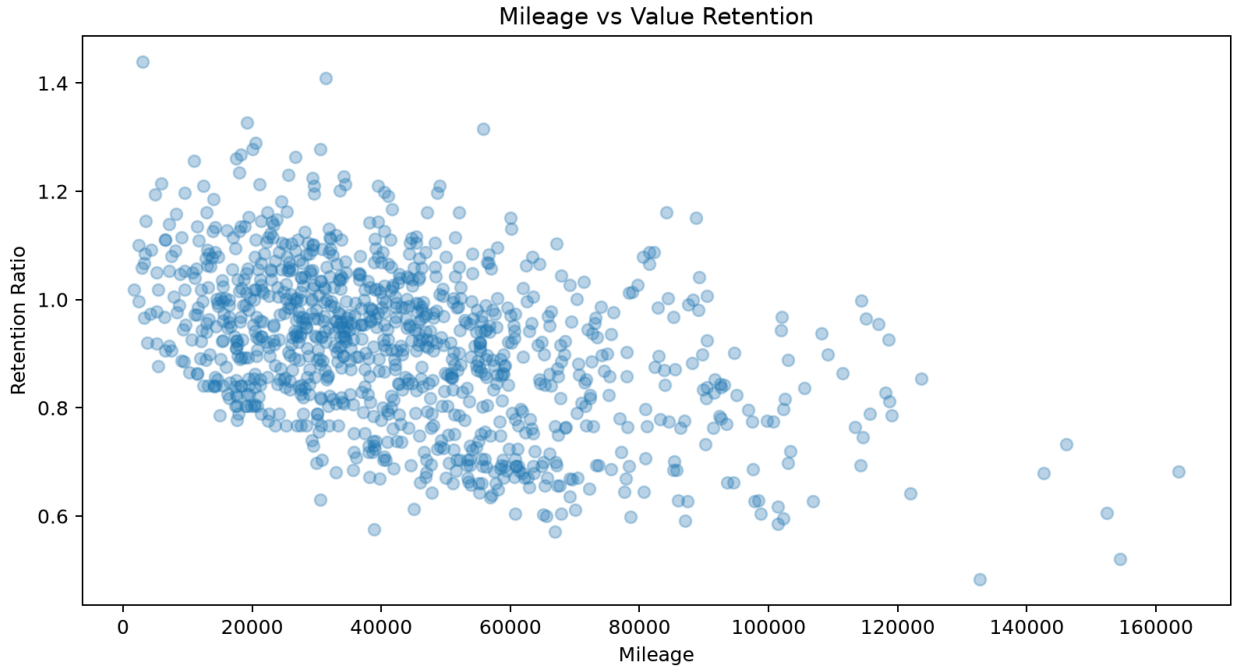


Figure 10: Mileage Versus Value Retention



These visualizations explore how vehicle age and mileage relate to the value retention ratio.The age-based visualization helps compare retention patterns across model years, while the mileage scatter plot provides a way to examine relationships between vehicle usage and retention.These

figures are useful for identifying patterns, but their interpretation must account for differences in make, model, trim, and MSRP reference values.

Conclusion

9.1 Summary of Project Findings

The project successfully developed a machine learning pipeline to predict used vehicle value retention using real world automotive listing data.A total of 1,832 vehicle listings were collected from Cars.com and Carvana, and 1,033 listings remained after MSRP matching and preprocessing. The project included three different manufacturers and nine models,three for each, and three regression algorithms were trained and evaluated based on them.Gradient Boosting Regression achieved the best performance, with an MAE of 0.0546, RMSE of 0.0724, and R^2of 0.7637.The feature importance analysis identified vehicle make, mileage, and model as the strongest predictors of the retention ratio. Mileage per year and vehicle age also contributed useful predictive information.Overall, the results demonstrated that machine-learning methods can identify meaningful patterns in used-vehicle pricing and value retention.

9.2 What I Learned

Through this project, I learned how to develop a complete machine-learning workflow, beginning with real-world data collection and ending with predictive modeling and result interpretation.Another important lesson was understanding that collecting data is only the beginning of a machine-learning project. Data cleaning, integration, and validation are essential because inaccurate or inconsistent information can affect the reliability of the final model. I also gained experience with web scraping, Pandas data processing, categorical encoding, feature engineering, and regression modeling using Scikit-Learn.Comparing multiple regression algorithms helped me understand the differences between linear models and nonlinear ensemble methods. I learned how MAE, RMSE, and R^2provide different perspectives on predictive performance.Most importantly I gained a lot more resilience.Finally, the feature importance analysis demonstrated how machine learning can be used not only to make predictions but also to investigate which variables contribute most to model performance.

9.3 Challenges Encountered

The first major challenge was collecting data from two websites with different structures. Cars.com and Carvana presented vehicle information differently, requiring separate scraping approaches and additional preprocessing to combine their records.The second biggest struggle was actually scrapping the data because it was facing security issues. After some research I found scrapfly which I have actually used before and utilized a webscraper API key to help bypass these problems.

The third major challenge was collecting and matching original MSRP information. Vehicle prices vary depending on model year, trim, drivetrain, and optional equipment. Matching exact original configurations was difficult because used listings did not always contain all the necessary information.To address this problem, standardized base-model MSRP references were used. Although this approach made matching more consistent, it introduced limitations when comparing higher-trim vehicles against base-trim prices.The final modeling dataset was also unevenly distributed across manufacturers, with substantially fewer Toyota listings than Honda and Nissan listings.

**** AI was used for debugging things like improper scrapping, setting up scrapfly API key to bypass anti scrapping measures, and data implementation. ****

MSRP reference:
https://usa.nissannews.com/en-US/releases/2023-nissan-rogue-pricing-starts-at-27760

https://www.onetoyota.com/inventory/used-2025-nissan-rogue-sv-awd-4d-sport-utility-jn8bt3bb7sw413375/

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https://www.carvana.com/?srsltid=AYCX9vJNWHxULV0DXZiBuzj3xQgaM_ShzSpNSlJsTuHYuLkvXXU933AN

https://scikit-learn.org/stable/